

# From Discriminative Scoring to Generative Creation: A Synthesis of Paradigms, Architectures, and Alignment Challenges in LLM-Based Recommendation

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The integration of Large Language Models (LLMs) into recommender systems marks a fundamental transition from traditional discriminative scoring to innovative generative frameworks [8]. While discriminative models struggle with data sparsity and semantic opacity, generative paradigms reconceptualize recommendation as sequence generation, leveraging world knowledge for zero-shot generalization [3]. However, this shift introduces a critical tension between linguistic semantics and collaborative filtering signals. To bridge this gap, recent research proposes novel tokenization strategies, such as semantic identifiers and order-agnostic sets, alongside hybrid architectures like LlamaRec that balance retrieval efficiency with generative reasoning [17]. Furthermore, frameworks like TCA4Rec establish explicit interfaces to align token-level generation with item-level collaborative objectives [148]. Despite these advancements, significant challenges persist, including hallucination risks, the "lost-in-the-middle" phenomenon in long-sequence modeling, and the computational latency of autoregressive decoding [58]. This document synthesizes current architectural innovations, evaluates the trade-offs between end-to-end generation and retrieval-augmented pipelines, and outlines future directions toward autonomous, controllable recommendation agents that harmonize semantic fluency with behavioral fidelity.

## Introduction: The Paradigm Shift to Generative Recommendation

The integration of Large Language Models (LLMs) into recommender systems signifies a fundamental transition from traditional discriminative scoring paradigms to innovative generative frameworks. Historically, recommender systems have relied on modular architectures comprising candidate generation, multi-stage ranking, and re-ranking, where models are trained separately using supervised objectives and hand-engineered features to estimate the probability of a user-item interaction [8]. These discriminative approaches, while effective in many domains, face persistent limitations including data sparsity, cold-start problems, and an inadequate semantic understanding of user and item content [51]. The emergence of LLMs offers a new paradigm that leverages vast world knowledge, logical reasoning capabilities, and natural language understanding to address these challenges, effectively bridging the gap between semantic content and collaborative signals [3].

Unlike conventional systems that calculate ranking scores for candidate items one by one, generative recommendation reconceptualizes the task as a sequence generation problem. In this framework, the model directly generates the target item or a list of recommendations autoregressively, utilizing the expressive power of LLMs to interpret context and learn user preferences [2]. This shift allows systems to move beyond opaque ID-based embeddings, which often lack interpretability, toward language-native mechanisms that can generalize across tasks, domains, and modalities [16]. By treating recommendation as a generation task, LLMs can incorporate external knowledge and contextual cues to operate effectively in zero-shot and few-shot scenarios, thereby mitigating the cold-start issues that plague traditional collaborative filtering methods [8].

The distinction between discriminative and generative paradigms is central to understanding this evolution. Discriminative LLMs for Recommendation (DLLM4Rec) typically utilize LLMs as feature extractors or rerankers within existing pipelines, whereas Generative LLMs for Recommendation (GLLM4Rec) employ the model as the core engine to produce recommendations directly [16]. Research indicates that while discriminative approaches benefit from the semantic richness of LLMs, they often fail to fully leverage the models' generative capacities. Conversely, generative approaches promise to simplify the recommendation pipeline into a single stage, eliminating the need for complex candidate retrieval and scoring modules [4]. However, this transition is not without challenges; existing generative methods must address issues such as the semantic gap between natural language and collaborative semantics, the efficiency of autoregressive generation, and the potential for hallucinations where the model generates invalid item identifiers [28].

To bridge the semantic-collaborative gap, recent methodologies have focused on aligning the language space of LLMs with the item space of recommendation systems. Approaches such as learning-based vector quantization and semantic identifier tokenization have been proposed to assign meaningful, non-conflicting IDs to items, enabling LLMs to integrate collaborative semantics with their inherent language understanding [137]. Furthermore, frameworks like TCA4Rec establish explicit optimization-level interfaces between collaborative filtering supervision and LLM generation, ensuring that behavioral signals explicitly regulate the generative process [148]. Despite these advancements, a tension remains between the rich semantic priors of LLMs and the statistical patterns inherent in user interaction data. Studies suggest that while LLMs possess robust semantic reasoning, they may struggle to assimilate statistical information such as item popularity solely from training samples, necessitating hybrid approaches that combine generative capabilities with traditional statistical signals [187].

The potential of this paradigm shift extends to addressing the limitations of data sparsity and enhancing personalization depth. By leveraging the extensive knowledge encoded in LLMs, generative models can infer user preferences from limited interaction history and utilize open-world knowledge to recommend items outside the immediate training distribution [97]. This capability is particularly valuable in long-tail scenarios, where traditional models often fail due to insufficient interaction data. Moreover, the generative nature of these systems facilitates explainable recommendations, as the model can naturally generate textual rationales alongside item suggestions, thereby enhancing user trust and transparency [276]. As the field progresses, the focus is shifting towards developing unified frameworks that can seamlessly integrate the reasoning abilities of LLMs with the precision of collaborative filtering, paving the way for next-generation recommender systems that are both semantically rich and behaviorally accurate [51].

## Historical Context and Limitations of Discriminative Models

The evolution of recommender systems has traversed a significant trajectory, moving from rudimentary heuristic-based approaches to sophisticated discriminative architectures grounded in deep learning. Initially, systems relied on deterministic rules and statistical aggregations, such as popularity-based metrics that ranked items by global interaction frequency or co-occurrence rules suggesting items based on pairwise co-interactions [8]. While these heuristic methods offered scalability and interpretability, they fundamentally lacked personalization, assuming user homogeneity and failing to adapt to individual contextual nuances [8]. This limitation spurred the adoption of Collaborative Filtering (CF), which operates on the homophily principle that users with similar historical interaction patterns share future preferences [8]. Despite its widespread success, CF and its matrix factorization derivatives struggle with data sparsity and the cold-start problem, where insufficient interaction history prevents accurate similarity computation [8]. Furthermore, traditional CF models often treat user-item interactions as static, ignoring the temporal dynamics inherent in human behavior [108].

To address the static nature of early models, the field shifted toward Sequential Recommender Systems (SRS), which model user-item interactions as dynamic sequences to capture evolving preferences and sequential dependencies [108]. Unlike conventional models that only capture general preferences, SRSs aim to understand the complex, multi-level cascading dependencies within a user's interaction history, such as the likelihood of purchasing a specific accessory after buying a primary device [108]. However, even these advanced discriminative models face persistent challenges. A primary limitation is the difficulty in capturing higher-order and long-term sequential dependencies; while Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks were introduced to model these sequences, they often suffer from overly strong order assumptions or generate false dependencies between adjacent items that are not semantically related [108]. Additionally, traditional SRS models typically rely on implicit feedback data, which is inherently incomplete and noisy [183]. For instance, standard models often ignore unclicked items presented to users, treating them as missing data rather than valuable negative signals that could refine the understanding of user disinterest [110]. This oversight leads to incomprehensive user representations and sub-optimal performance in capturing dynamic interest shifts [110].

Another critical deficiency in discriminative models is their inability to adequately handle long-tail items and the resulting popularity bias. Traditional systems, driven by optimization objectives that favor frequent interactions, disproportionately recommend popular items, thereby overshadowing niche but potentially relevant content [121]. This bias is exacerbated in long-tail scenarios where interaction data is scarce, causing models to fail in generalizing to less popular items or new users [61]. The reliance on ID-based representations further limits these models, as they struggle to incorporate rich semantic information from item descriptions or user reviews, leading to a lack of semantic understanding and explainability [8]. While some approaches have attempted to integrate content features through hybrid architectures, they often fail to fully leverage the world knowledge and reasoning capabilities necessary to interpret complex user intents or generate diverse recommendations beyond historical patterns [125]. Moreover, discriminative models are often constrained by their modular architectures, where candidate generation, ranking, and re-ranking are trained separately with supervised objectives [8]. This fragmentation limits the system's ability to perform holistic reasoning across the entire recommendation pipeline [8]. The inability to naturally integrate multi-modal data or handle flexible interaction orders further restricts their applicability in real-world scenarios where user behaviors are not strictly sequential but may involve flexible, collective dependencies [108]. Consequently, while discriminative models have achieved considerable success in optimizing accuracy metrics on dense datasets, they remain ill-equipped to address the nuanced requirements of dynamic intent prediction, serendipitous discovery, and robust handling of sparse or noisy data distributions [195].

## The Emergence of the Generative Paradigm and Taxonomy

The integration of Large Language Models (LLMs) into recommender systems has catalyzed a fundamental paradigm shift, moving the field away from traditional discriminative frameworks toward a generative recommendation paradigm. Historically, recommender systems have operated on a retrieval-based model where the core task involves calculating ranking scores for a pre-defined set of candidate items to identify the most relevant ones for a user [4]. This conventional approach, while effective in many contexts, is inherently limited by the static nature of the item corpus and the reliance on implicit feedback mechanisms such as clicks, which often fail to capture the full spectrum of user intent [195]. In contrast, the emerging generative paradigm reconceptualizes recommendation as a text generation task. Instead of scoring existing candidates, LLMs are employed to directly generate target items, item identifiers, or entire interaction sequences, thereby leveraging the models' inherent capabilities in sequence modeling, reasoning, and natural language understanding [3].

This transition represents more than a mere architectural adjustment; it signifies a redefinition of the recommendation objective itself. Traditional discriminative models estimate the probability of a label given an observation, formally expressed as  $P(Y|X)$ , whereas generative models estimate the distribution of observations given a label, or  $P(X|Y)$ , allowing for the creation of rich, complex, and structured outputs [92]. By treating recommendation as a generation problem, systems can move beyond simply retrieving human-generated content to potentially repurposing existing items or creating entirely new personalized content that satisfies specific, diverse information needs [195]. This capability addresses critical limitations of retrieval-based systems, particularly the insufficiency of existing corpora to meet niche user preferences and the inefficiency of passive feedback loops. The generative approach enables users to express needs via natural language instructions, facilitating a more interactive and precise alignment between user intent and system output [195].

A central challenge and area of innovation within this paradigm is the mechanism by which items are represented and generated. Early implementations often treated LLMs as feature extractors within conventional pipelines, failing to fully exploit their generative potential [4]. Contemporary research has since developed specialized frameworks where items are tokenized into semantic identifiers, allowing the LLM to autoregressively generate sequences of tokens that correspond to recommended items [138]. Various strategies have been proposed to bridge the gap between the language space and the item space. For instance, some approaches utilize a bi-step grounding paradigm where the model first generates meaningful tokens in a recommendation space and then grounds these tokens to actual items in the corpus, incorporating statistical information like popularity to refine the final selection [187]. Others advocate for order-agnostic identifiers to mitigate the inefficiencies of step-by-step generation and the local optima problems associated with beam search in token-sequence identifiers [132].

Despite the promise of direct generation, significant contradictions and challenges remain regarding the fidelity of item-level preference modeling. Theoretical analyses suggest that approximating item-level distributions via token-level generation, as done in standard autoregressive models, introduces systematic biases due to beam search pruning or assumptions of token independence [58]. To address this, alternative frameworks like SimGR propose modeling item-level preference distributions directly in a shared latent space, arguing that token-level generation cannot faithfully substitute for item-level generation without inducing distributional distortion [58]. Furthermore, the integration of collaborative filtering (CF) signals into generative models remains non-trivial. While CF relies on item-level preference modeling, LLMs optimize for token-level next-token prediction, creating a fundamental mismatch. Recent works like TCA4Rec attempt to resolve this by establishing explicit optimization-level interfaces that align collaborative signals with LLM generation through token-level collaborative alignment [148].

The taxonomy of generative recommendation is further diversified by the scope of generation, ranging from generating item identifiers to generating full content or queries. In domains like music or video, the paradigm extends to generating multimodal content or repurposing existing media based on user instructions, effectively turning the recommender into a creator [195]. In search-centric scenarios, the generative paradigm manifests as query generation, where LLMs produce personalized queries to retrieve relevant items from a large pool, bypassing the need for candidate pre-selection and mitigating issues like the "lost in the middle" phenomenon [224]. Additionally, unified frameworks are emerging that treat both search and recommendation as sequential generation tasks, utilizing dual-purpose identifiers to balance semantic relevance and collaborative signals [265].

The evolution of this paradigm also encompasses the methodology of adapting LLMs to recommendation tasks. Approaches vary from zero-shot prompting, which leverages the pre-trained world knowledge of LLMs but often struggles with domain-specific nuances [89], to extensive fine-tuning strategies that align LLMs with recommendation data. Frameworks such as P5 unify various recommendation tasks into a "Pretrain, Personalized Prompt, and Predict" paradigm, treating all data as natural language sequences to enable instruction-based recommendations [49]. Similarly, TALLRec demonstrates that efficient tuning with limited data can significantly enhance recommendation capabilities, bridging the gap between general language understanding and specific recommendation objectives [178]. However, the trade-off between maintaining the general reasoning capabilities of LLMs and overfitting to specific recommendation patterns remains a critical consideration, with some reinforcement learning approaches like REC-R1 aiming to optimize generation without catastrophic forgetting of general skills [56].

Ultimately, the generative paradigm offers a pathway to overcome the data sparsity, cold-start problems, and lack of explainability that plague traditional systems by leveraging the vast knowledge encoded in LLMs [8]. By shifting from scoring to generating, these systems promise a more flexible, interactive, and semantically rich future for personalized information access, although realizing this potential requires resolving ongoing challenges in grounding, efficiency, and the faithful modeling of user preferences [4].

## Architectural Frameworks for Generative Recommendation

The architectural landscape of generative recommendation systems has evolved from simple prompt-based interfaces to complex, multi-stage pipelines that integrate Large Language Models (LLMs) with traditional retrieval mechanisms. A fundamental divergence exists between end-to-end generative models, which treat recommendation as a direct sequence generation task, and hybrid architectures that leverage LLMs for specific components such as ranking, data augmentation, or semantic indexing. The end-to-end paradigm aims to simplify the recommendation process by directly generating item identifiers or descriptions from the complete pool of candidates, thereby bypassing the traditional multi-stage pipeline of scoring and re-ranking [4]. This approach posits that LLMs can fully leverage their generative power to model user preferences and item characteristics simultaneously, moving beyond the limitations of discriminative models that estimate probabilities based on fixed feature sets [92]. However, scaling these pure generative approaches to industrial environments presents significant challenges regarding latency and the handling of massive candidate sets, leading to the development of more nuanced architectural solutions.

To address the scalability and efficiency constraints of autoregressive generation, several frameworks propose hybrid multi-stage pipelines that combine the semantic reasoning of LLMs with the speed of traditional retrievers. For instance, the LlamaRec framework introduces a two-stage architecture where a small-scale sequential recommender first retrieves a candidate set, which is then ranked by an LLM using a verbalizer-based approach rather than generating long text sequences [17]. Similarly, the TalkPlay-Tools system positions the LLM as a central orchestrator that plans and executes tool calls to specialized components, including boolean filters, sparse retrieval, and dense retrieval modules, effectively creating a unified retrieval-reranking pipeline [86]. This modular design allows the system to exploit the strengths of different retrieval methods while maintaining the flexibility of natural language interaction. Furthermore, the GeneRec paradigm extends this concept by introducing an AI generator loop that can repurpose existing items or create new content based on user instructions, complementing the traditional retrieval-based loop with generative capabilities [195].

A critical architectural component in many generative systems is the method used to represent items, as standard tokenization often fails to capture collaborative filtering signals or leads to inefficient decoding. To mitigate the "lost in the middle" phenomenon and improve generation efficiency, researchers have developed novel item indexing strategies. The SETRec model proposes an order-agnostic identifier paradigm that represents items as sets of tokens, eliminating token dependency and enabling simultaneous generation [132]. In contrast, the PLUM framework utilizes Semantic IDs derived from modality encoders, employing continued pre-training and task-specific fine-tuning to adapt LLMs for industry-scale generative retrieval [68]. Despite the popularity of Semantic IDs, recent analyses suggest that this approach may face scaling bottlenecks, with performance saturating as model size increases, whereas the "LLM-as-RS" paradigm, which relies on direct language modeling of item attributes, demonstrates superior scaling properties [171]. Consequently, frameworks like TokenRec have emerged to quantize masked user and item representations from collaborative filtering into discrete tokens, ensuring that high-order collaborative knowledge is seamlessly incorporated into the LLM's token space [41].

Addressing the limitation of LLM context windows in modeling long-term user behavior has led to the integration of memory and retrieval-augmented generation (RAG) mechanisms within these architectures. The AutoMR framework introduces an automatic memory-retrieval system that stores long-term user interests and extracts relevant information dynamically, overcoming the restriction of focusing only on recent interactions [13]. Similarly, the K-RagRec approach enhances recommendations by retrieving structured information from knowledge graphs to augment the LLM's generation process, thereby reducing hallucinations and incorporating up-to-date domain knowledge [253]. Distinct from user-based RAG, the ItemRAG method retrieves relevant items based on item-item co-purchase histories, allowing the LLM to capture complex co-purchase patterns that are beneficial for cold-start scenarios [133]. These retrieval-enhanced architectures highlight a shift towards systems that do not rely solely on the parametric knowledge of the LLM but actively interface with external data sources to ground their recommendations.

Furthermore, the tension between the semantic capabilities of LLMs and the collaborative signals inherent in traditional recommendation data has spurred architectures designed for explicit alignment. The TCA4Rec framework establishes a token-level collaborative alignment mechanism that projects item-level collaborative filtering logits into the LLM's token space, enabling explicit optimization of collaborative signals during generative training [148]. In a similar vein,

the CLLM4Rec model extends the LLM vocabulary with user and item ID tokens and employs a soft plus hard prompting strategy to tightly integrate collaborative and content semantics [28]. Conversely, some approaches argue that direct fine-tuning on recommendation data is insufficient due to the mismatch between linguistic and collaborative semantics, proposing instead frameworks like EAGER-LLM that integrate exogenous behavioral information through dual-source item indices and non-invasive alignment tasks [33]. The diversity of these architectural choices underscores the ongoing trade-offs between accuracy, scalability, and the depth of semantic understanding, with no single paradigm yet dominating the field. Instead, the trend points toward increasingly sophisticated hybrid systems that strategically deploy LLMs where their reasoning and generative capabilities offer the most distinct advantage over traditional methods [8].

## End-to-End Generative Models and Tokenization

The paradigm of end-to-end generative recommendation represents a fundamental shift from traditional discriminative architectures, which typically rely on multi-stage pipelines involving candidate retrieval, scoring, and re-ranking. Instead, end-to-end models aim to simplify this process into a single stage where a Large Language Model (LLM) directly generates the target item or sequence based on user history, effectively bypassing the need for explicit candidate scoring against a fixed pool [4]. This approach reconceptualizes the recommendation problem as a generation task, leveraging the autoregressive capabilities of LLMs to predict the next item token by token, similar to next-word prediction in natural language processing [27]. By treating user interaction sequences as a distinct language, these models can theoretically access the vast world knowledge encoded within pre-trained LLMs, offering advantages in semantic understanding and reasoning that traditional ID-based collaborative filtering methods often lack [126].

A critical component enabling this generative paradigm is the tokenization strategy, specifically how items are represented within the LLM's vocabulary. Traditional recommendation systems rely on discrete, learnable ID embeddings, but integrating these directly into LLMs presents challenges due to the mismatch between the semantic space of language and the collaborative space of item IDs. To address this, various indexing and tokenization methods have been proposed. For instance, some frameworks utilize random, sequential, or collaborative indexing to map item IDs to text tokens, allowing the LLM to generate items as text sequences [274]. However, purely ID-based generation can suffer from hallucinations, where the model generates tokens that do not correspond to any real item in the catalog. To mitigate this, researchers have introduced grounding mechanisms. The BIGRec framework, for example, employs a bi-step grounding paradigm where the LLM first generates tokens in a "recommendation space" (which may include hypothetical items) and then maps these generated tokens to actual items in the "actual item space" using retrieval or similarity metrics [187]. This two-step process ensures that the generative flexibility of the LLM is retained while guaranteeing the validity of the final recommendation.

Despite the theoretical elegance of direct generation, practical implementation faces significant hurdles, particularly regarding efficiency and the handling of large candidate pools. Generating items one by one via autoregressive decoding can be computationally expensive and slow, making it less suitable for real-time industrial applications compared to the rapid scoring of discriminative models. Consequently, hybrid approaches have emerged that blend generative capabilities with efficient ranking mechanisms. LlamaRec proposes a two-stage framework where a small-scale sequential recommender retrieves a candidate set, and the LLM is then used to rank these candidates. Crucially, instead of generating long text descriptions, LlamaRec adopts a verbalizer-based approach that transforms output logits into probability distributions over the candidate items, significantly improving inference speed [17]. Similarly, other works suggest using LLMs to generate personalized queries to retrieve relevant items from the entire pool, thereby eliminating the need for pre-selecting candidates while avoiding the inefficiency of prompting the LLM with thousands of items [224].

The effectiveness of end-to-end generative models also depends heavily on how well they capture sequential dependencies and user intent. While LLMs possess strong context-awareness, studies indicate that they do not always fully understand the sequential information inherent in user interaction histories when applied directly to recommendation tasks [145]. To enhance this capability, methods like CALRec introduce contrastive alignment losses alongside language modeling objectives, fine-tuning the LLM to better align user behavior sequences with item representations [87]. Furthermore, the integration of collaborative signals remains a challenge; while LLMs excel at semantic reasoning, they often struggle to capture the statistical patterns of user-item interactions that define collaborative filtering. Approaches such as EAGER-LLM attempt to bridge this gap by integrating exogenous behavioral signals with endogenous semantic knowledge through dual-source item indices and non-invasive alignment tasks [33].

Contradictions and debates persist regarding the optimal role of the LLM in the recommendation pipeline. Some research argues that fine-tuning large models is essential to inject domain-specific recommendation knowledge, as pre-trained LLMs lack the specific collaborative patterns required for accurate prediction [178]. Conversely, other studies suggest that LLMs can serve effectively as zero-shot or few-shot recommenders by leveraging their general reasoning abilities, provided that the prompting strategy and item representation are carefully designed [89]. Additionally, there is ongoing discussion about whether generative models can truly replace discriminative scorers in

all scenarios. While generative models offer superior explainability and the ability to handle cold-start items through semantic understanding, they may still lag behind highly optimized discriminative models in pure accuracy metrics for dense interaction data [164]. The field is thus moving towards unified frameworks that can dynamically leverage the strengths of both generative and discriminative approaches, utilizing tokenization strategies that preserve both semantic richness and collaborative fidelity.

## Retrieval-Augmented and Multi-Stage Architectures

To address the computational inefficiency and context limitations inherent in pure generative models, the field of generative recommendation has increasingly adopted retrieval-augmented generation (RAG) and multi-stage architectures. These frameworks typically decouple the process into a retrieval phase, where a traditional or specialized retriever identifies a manageable candidate set, followed by a generation or ranking phase where a Large Language Model (LLM) processes these candidates for final selection. This hybrid approach mitigates the "lost in the middle" phenomenon and the prohibitive latency associated with feeding massive item catalogs directly into LLM prompts [224]. By leveraging the efficiency of dense retrieval or heuristic matching for candidate generation, these systems preserve the semantic reasoning and world knowledge capabilities of LLMs for the more critical task of re-ranking and personalization [17].

A prominent strategy within this paradigm involves using LLMs to generate queries or semantic representations that drive the retrieval process, rather than generating items directly from the full vocabulary. For instance, a framework has been proposed where the LLM generates hypothetical search queries based on a user's interaction history, which are then used to retrieve items via a traditional search engine like BM25 [101]. Similarly, another approach combines generative language models with dense semantic search, moving beyond lexical matching to retrieve relevant items for sequential recommendation, thereby achieving state-of-the-art performance on various datasets [37]. This query-generation approach effectively bridges the gap between collaborative filtering signals and semantic understanding, allowing the system to handle cold-start scenarios and diverse user intents more robustly [120]. Furthermore, recent advancements have explored using LLMs as universal retrievers capable of handling multiple objectives simultaneously. A Universal Retrieval Model demonstrates that LLMs can adaptively generate candidate sets from tens of millions of items with low latency by employing matrix decomposition and probabilistic sampling, outperforming expert models designed for specific retrieval tasks [201].

Beyond simple query generation, multi-stage architectures often incorporate sophisticated memory and knowledge retrieval mechanisms to enhance the context available to the generative model. The limited context window of LLMs often restricts them to recent interactions, neglecting long-term user interests. To counteract this, frameworks introduce automatic memory-retrieval systems that store long-term interests in an external memory and extract relevant information dynamically during the generation process [13]. In parallel, knowledge graph-based retrieval has been integrated to provide structured, up-to-date domain knowledge, addressing issues of hallucination and knowledge cutoffs. One such approach retrieves high-quality structural information from knowledge graphs to augment the recommendation generation process, ensuring that the LLM's output is grounded in factual relationships [253]. Item-based retrieval strategies further refine this by retrieving relevant items based on co-purchase histories rather than user similarities, which proves particularly effective for cold-start item recommendations [133].

The alignment between the retrieval and generation stages remains a critical challenge in these architectures. Misalignment can lead to suboptimal performance where the retrieved candidates do not match the generative model's reasoning capabilities. To address this, some frameworks employ reinforcement learning to create a collaborative feedback loop, iteratively updating the retriever based on the LLM's feedback to ensure better candidate sets [215]. Additionally, hybrid retrieval mechanisms that mine both ID-based user patterns and text-based item features have been proposed to overcome data sparsity and heterogeneity, particularly in federated learning settings [185]. The effectiveness of these multi-stage approaches is further validated by systems that utilize small-scale sequential recommenders for initial candidate retrieval before feeding the results to an LLM for verbalizer-based ranking, achieving superior efficiency and performance compared to end-to-end generative methods [17].

Contradictions and nuances exist regarding the optimal balance between retrieval dependence and generative autonomy. While some studies argue that heavy reliance on retrieval limits the creative potential of generative models, others contend that without retrieval, LLMs struggle to capture collaborative filtering signals effectively [164]. The debate extends to the modeling of item distributions, where pure autoregressive generation may introduce systematic biases due to token-level approximations. Frameworks propose directly modeling item-level preference distributions in a shared latent space to mitigate these distortions, suggesting that future multi-stage architectures may need to evolve beyond simple retrieve-then-rank pipelines to more integrated modeling approaches [58]. Nevertheless, the consensus indicates that retrieval-augmented and multi-stage architectures currently offer the most viable path for deploying

LLMs in industrial-scale recommendation systems, balancing the trade-offs between accuracy, scalability, and real-time performance [8].

## Hybrid Collaboration and Parameter-Efficient Fine-Tuning

The integration of Large Language Models (LLMs) into recommendation systems has necessitated a shift from purely discriminative scoring to generative and collaborative frameworks, yet the prohibitive computational costs of LLM inference and training have driven the development of hybrid architectures. These architectures synergize the reasoning capabilities of LLMs with the efficiency of Small Recommendation Models (SRMs) to address industrial latency constraints. A prominent strategy involves device-cloud collaboration, where complex reasoning and candidate generation are offloaded to cloud-based LLMs while real-time updates and reranking occur on edge devices via SRMs. This approach allows the system to balance freshness with computational economy by triggering cloud requests only when local SRM rankings diverge significantly from cloud-based LLM outputs, ensuring consistency without constant high-latency communication [25]. Parallel to this spatial decoupling, other frameworks adopt a temporal decoupling strategy where LLMs are utilized offline to extract open-world knowledge about users and items. This knowledge is condensed into augmented vectors that enhance conventional recommendation models during online serving, effectively bridging the gap between closed-loop interaction data and open-world semantics without incurring online inference latency [40]. Similarly, data-centric hybrid approaches leverage LLMs to synthesize and filter interaction data to boost small Item-to-Item models, avoiding the need to alter the online serving model architecture while mitigating data sparsity [85].

While hybrid architectures address deployment efficiency, adapting LLMs to specific recommendation tasks often requires fine-tuning, leading to significant research in Parameter-Efficient Fine-Tuning (PEFT). Although Low-Rank Adaptation (LoRA) is widely adopted, standard implementations often fail to capture personalized user characteristics because the adaptation matrix is typically shared across all users, leading to suboptimal performance in heterogeneous behavioral contexts [298]. To resolve this, recent studies propose personalized LoRA modules that maintain independent adaptation parameters for different users, significantly improving performance while maintaining efficiency [298]. Furthermore, instance-wise LoRA approaches treat sequential recommendation as a multi-task learning problem, integrating LoRA with a Mixture of Experts framework to dynamically adjust parameters for specific user sequences, thus preventing negative transfer between disparate behavioral patterns [102]. Despite the popularity of PEFT, empirical analyses suggest that fine-tuning all parameters of an LLM can still yield superior recommendation performance compared to parameter-efficient methods, indicating a persistent trade-off between efficiency and efficacy that requires careful consideration based on resource availability [103].

The synergy between hybrid collaboration and fine-tuning is further evident in efforts to bridge the semantic gap between natural language and collaborative signals. Some models tightly integrate the LLM paradigm with the ID paradigm by extending the vocabulary with user and item ID tokens, employing soft and hard prompting strategies to learn collaborative semantics efficiently [28]. This aligns with findings that LLMs often struggle to capture collaborative patterns inherent in traditional matrix factorization models unless explicitly grounded in structured interaction data [164]. Consequently, retrieval-augmented generation has emerged as a critical mechanism, where LLMs are grounded via retrieved collaborative evidence to align their reasoning with dataset-specific user-item interaction patterns, particularly improving long-tail recommendation performance [45]. Additionally, efficient tuning frameworks have demonstrated that even with limited datasets, strategic alignment can enable LLMs to achieve robust cross-domain generalization, challenging the notion that massive data is always prerequisite for effective adaptation [178]. Contradictions remain regarding the optimal role of LLMs; while some advocate for direct generative recommendation, others argue LLMs are better suited as enhancers due to hallucination risks and the "lost in the middle" phenomenon when handling large candidate pools, suggesting a future where LLMs and SRMs complement each other rather than one replacing the other [224].

## Modeling User Behavior and Sequential Dynamics

A critical challenge in the evolution of generative recommendation systems lies in the effective modeling of user behavior, particularly when confronting the complexities of long, heterogeneous interaction histories. While Large Language Models (LLMs) offer unprecedented reasoning capabilities and world knowledge, they frequently suffer from a phenomenon termed "lifelong sequential behavior incomprehension," where performance degrades as the length of the user interaction sequence increases, even when the total token count remains well within the model's theoretical context window limits [24]. Traditional sequential models have historically struggled with capturing long-term dependencies and higher-order sequential patterns, often failing to distinguish between transient noise and enduring user interests [Sequential Recommender Systems: Challenges, Progress and Prospects]. The integration of LLMs was anticipated to resolve these issues through superior semantic understanding; however, empirical evidence suggests that LLMs often exhibit a "lost-in-the-middle" effect, where critical information embedded in the middle of long sequences is overlooked, leading to biased or incomplete user profiling [54].

To mitigate the limitations imposed by context length and information heterogeneity, recent research has increasingly turned toward retrieval-augmented strategies and memory mechanisms. Instead of feeding raw, exhaustive interaction logs directly into the LLM, frameworks like AutoMR propose an Automatic Memory-Retrieval approach that stores long-term interests in an external memory bank and retrieves only the most relevant segments for next-item generation [13]. Similarly, the ReLLa framework addresses the incomprehension problem through a full-stack optimization strategy involving Semantic User Behavior Retrieval (SUBR), which selects semantically relevant behaviors rather than merely the most recent ones, thereby reducing sequence heterogeneity and enhancing the LLM's ability to extract essential knowledge [5]. These retrieval-based methods are complemented by memory-assisted approaches like MAP, which constructs dynamic history profiles and extracts relevant memories based on similarity to enhance personalized recommendations without incurring the computational cost of processing entire histories at every inference step [130]. Furthermore, the LIBER framework introduces a streaming partition paradigm to condense lengthy behavior sequences into manageable chunks, leveraging cascading LLM calls to infer insights incrementally while maintaining computational efficiency [200].

Beyond simple retrieval, sophisticated architectural innovations have been proposed to better capture the dynamic evolution of user interests within long sequences. The HiT-LBM framework employs a Hierarchical Tree Search mechanism combined with Chunked User Behavior Extraction to navigate lifelong behavior data, utilizing process rating models to ensure the quality of interest extraction at each step and prevent error accumulation [218]. This aligns with findings that direct fine-tuning of LLMs on long sequences often fails to capture the nuanced temporal shifts in user preferences, necessitating specialized modules for interest fusion and evolution modeling [200]. In the realm of cross-domain and heterogeneous behaviors, the HUM method introduces compression and robustness enhancers to effectively model user preferences across multiple domains, addressing the "domain seesaw" phenomenon where optimization in one domain degrades performance in another [204]. Additionally, the PersonaX framework decouples user profiling from online inference by performing offline multi-persona modeling on clustered behavior segments, allowing for the retrieval of high-quality, diverse user personas that capture both prototypical and unique interests without real-time latency overhead [54].

Contradictions and debates persist regarding the optimal method for aligning LLMs with sequential data. While some approaches advocate for pure text-based inputs where item IDs are replaced by titles to leverage semantic knowledge [39], others argue that this discards valuable collaborative filtering signals inherent in ID-based representations. To bridge this gap, hybrid models like LLaRA and CALRec integrate ID-based embeddings from traditional recommenders with textual features, using projectors to align these distinct modalities within the LLM's input space [146; 87]. The CALRec framework further demonstrates that a two-stage fine-tuning paradigm, combining multi-domain joint training with target-specific adaptation and contrastive losses, significantly outperforms single-stage approaches, suggesting that the alignment of behavioral knowledge requires structured optimization rather than simple prompt engineering [87]. Moreover, the effectiveness of generative decoding itself is under scrutiny; SimGR highlights that standard autoregressive generation introduces systematic biases due to token-level approximation errors, proposing instead a framework that directly models item-level preference distributions to avoid distributional distortion [58].

The handling of noise and unclicked behaviors also represents a significant frontier in sequential modeling. Traditional models often ignore unclicked items, yet frameworks like XDM demonstrate that incorporating unclicked exposure data via triplet metric learning provides a more comprehensive view of user preferences by treating unclicked items as intermediate feedback between clicked items and random negatives [110]. In federated settings where data is sparse and noisy, the LUMOS framework leverages on-device LLMs to synthesize diverse behavioral trajectories, including counterfactuals, to enrich contrastive learning without compromising privacy [289]. Finally, the emergence of agent-based simulators like DyTA4Rec and Lusifer indicates a shift toward using LLMs not just for prediction but for simulating dynamic, temporal-aware user feedback, enabling the training of robust recommenders in environments that mimic the fluid nature of real-world user interests [82; 167]. These diverse approaches collectively underscore that effective sequential dynamics modeling requires a synthesis of retrieval, memory, hybrid alignment, and noise-robust architectures to fully harness the potential of LLMs in generative recommendation.

## Addressing Long-Sequence Incomprehension and Memory Mechanisms

The integration of Large Language Models (LLMs) into sequential recommendation has exposed a critical paradox known as "lifelong sequential behavior incomprehension," where models fail to extract meaningful patterns from long user histories even when token counts remain well within context limits [5]. This phenomenon manifests as a performance peak at relatively short sequence lengths, followed by a degradation as history extends, indicating that raw textual context alone is insufficient for capturing dynamic user interests [24]. To resolve this, recent research has pivoted toward retrieval-based strategies and sophisticated memory architectures that curate high-quality behavioral data and align LLM representations with collaborative filtering signals.

A foundational approach to mitigating sequence incomprehension is Semantic User Behavior Retrieval (SUBR), which replaces chronological truncation with the selection of semantically relevant behaviors. By retrieving the top-K behaviors most relevant to a target item, this strategy reduces the heterogeneity of the input sequence, thereby lowering the cognitive load on the LLM and facilitating better knowledge extraction [5]. This retrieval paradigm is further enhanced by injecting collaborative knowledge directly into the prompt through soft prompt augmentation, ensuring that item representations align with recommendation-specific semantics rather than relying solely on textual descriptions [24]. Furthermore, relying exclusively on textual semantics has been deemed insufficient; consequently, frameworks have been developed to retrieve collaborative information in natural language form, employing self-assessing mechanisms to prioritize the most informative user behaviors for guiding the LLM [115].

Beyond simple retrieval, advanced memory architectures have been engineered to manage the redundancy and noise inherent in lifelong behavior logs through incremental and hierarchical processing. The LIBER framework introduces a streaming partition mechanism that condenses lengthy sequences into shorter, manageable partitions in an incremental paradigm, allowing LLMs to infer insights cascadingly while mitigating computational costs [200]. Complementing this partitioning strategy, the HiT-LBM framework employs a hierarchical tree search to navigate chunked user behaviors, utilizing rating models to select optimal interest paths and fuse temporal information, thus addressing the "lost-in-the-middle" phenomenon where critical mid-sequence context is overlooked [218]. In a different approach to memory management, PersonaX decouples user profiling from online inference by clustering historical behaviors and selecting sub-sequences that balance prototypicality and diversity, generating cached textual personas that serve as efficient memory for downstream agents [54]. These memory-enhanced approaches are also crucial for conversational systems, where frameworks utilize entity-based memory banks to refine preferences and reduce noise across historical dialogue sessions [131].

The alignment of LLM representations with collaborative signals represents another critical frontier, ensuring that retrieved data is interpreted correctly through parameter-efficient tuning and prompt engineering. AdaptRec addresses the challenge of transforming collaborative signals into a format LLMs can understand by employing a self-adaptive user selection mechanism and constructing user-contextualized prompts that explicitly integrate similar user behaviors [191]. In parallel, methods like MoRE introduce a mixture of reflectors to decouple explicit features from implicit behavioral patterns and capture cross-user collaborative filtering signals, ensuring that the LLM understands latent preference structures rather than relying on surface-level text [160]. Additionally, the integration of unclicked behaviors into sequence modeling provides a more comprehensive view of user preferences by treating unclicked items as intermediate feedback, a signal often lost in standard retrieval mechanisms that focus only on positive interactions [110].

Despite these advancements, contradictions remain regarding the efficiency and generalizability of these memory mechanisms, particularly concerning the trade-off between static retrieval and dynamic temporal modeling. While some studies advocate for full fine-tuning with specialized LoRA variants to capture long-sequence information [24], others argue for parameter-efficient transfer learning or frozen backbones with lightweight adapters to maintain scalability [118]. Furthermore, the reliance on static retrieval introduces latency concerns, prompting frameworks like DyTA4Rec to focus on dynamic temporal-aware agents that update profiles in real-time rather than relying solely on static retrieval [82]. The tension between capturing global collaborative patterns and maintaining personalized, temporal dynamics suggests that future solutions must hybridize retrieval-based data curation with adaptive memory structures that evolve alongside user interests, ensuring that the "incomprehension" of long sequences is resolved not just by shortening them, but by enriching their semantic and collaborative density.

## Capturing Dynamic User Intent and Multi-Interest Modeling

User preferences are inherently non-static, characterized by fluid shifts in intent and the coexistence of diverse interests that traditional collaborative filtering methods often fail to capture. To address the limitations of static modeling, recent research has increasingly turned to Large Language Models (LLMs) to interpret the semantic nuances of user behavior sequences. A primary advancement in this domain is the integration of sequential deep learning architectures with LLMs to dynamically predict intent. For instance, the DUIP framework combines Long Short-Term Memory (LSTM) networks with LLMs, where the LSTM models temporal dependencies to generate dynamic prompts that guide the LLM in predicting the next item of interest, effectively handling real-time intent adaptation and cold-start scenarios [6]. Similarly, other approaches leverage LLMs to enrich oversimplified log records in smart spaces by constructing graphs of individual behavior preferences that transform into soft continuous prompts, thereby capturing personalized preferences that standard sequential models miss [79].

A significant challenge in modeling dynamic intent lies in processing lifelong user behavior sequences, which are often too lengthy for standard LLM context windows and contain irregularly fluctuating interests. To mitigate computational overhead while capturing these dynamics, the LIBER framework proposes a lifelong user behavior modeling approach that partitions lengthy sequences into shorter segments, utilizes LLMs in a cascading manner to infer insights from these partitions, and fuses the textual outputs into a comprehensive representation [200]. Complementing this, the HiT-LBM framework employs a hierarchical tree search strategy combined with chunked behavior extraction to capture diverse interests and their evolution over time, ensuring that the quality of interest generation does not degrade across long sequences [218]. Furthermore, the PersonaX framework addresses the inefficiency of online profiling by segmenting historical behaviors into clustered groups and performing offline multi-persona profiling, which allows for the retrieval of fine-grained, cached textual personas that capture diverse user interests without incurring real-time latency [54].

Beyond handling sequence length, accurately disentangling multiple concurrent user interests remains a critical objective. Traditional methods often rely on heuristic assumptions where co-occurring items indicate a single interest, which can lead to semantic misalignment. The LDMI framework addresses this by employing an LLM-driven dual-level multi-interest modeling approach; at the individual level, it allocates items into semantic clusters, and at the crowd level, it synthesizes users with rich behaviors to perform contrastive learning, thereby refining the granularity of interest representation [31]. In parallel, the MoRE framework introduces a mixture of reflectors to decouple explicit features from implicit behavioral patterns and capture cross-user collaborative signals, adapting to evolving preferences through a dynamic selection mechanism [160]. Additionally, the SUM model utilizes a multi-channel network with an erase-and-add paradigm to maintain and update user states across different interest channels, offering an efficient solution for large-scale sequential recommendations [261].

The semantic grounding of user intent is further enhanced by integrating diffusion models and generative retrieval techniques. The LLMDiRec approach integrates LLM-based semantic embeddings into an intent-aware diffusion model, using dynamic fusion mechanisms to align collaborative signals with rich semantic representations, which proves particularly effective for long-tail items and cold-start users [290]. Similarly, generative retrieval frameworks like Mender condition recommendation systems on explicitly generated user preferences derived from reviews, enabling the system to be steered by natural language instructions during inference [19]. The GLIDE system at Spotify further demonstrates the efficacy of generative retrieval by formulating recommendation as an instruction-following task over a discretized catalog, injecting long-term user embeddings as soft prompts to balance stable preferences with changing intent [207].

Contradictions and agreements exist regarding the optimal method for integrating collaborative signals with semantic understanding. While some studies argue that fine-tuning LLMs on interaction data is necessary to bridge the gap between language and recommendation spaces [126], others suggest that parameter-efficient strategies, such as instance-wise LoRA or collaborative retrieval-augmented generation, can achieve superior alignment without the prohibitive costs of full fine-tuning [102; 45]. Moreover, the role of simulation in understanding dynamic intent is gaining traction; frameworks like Lusifer and SimUSER utilize LLMs to generate dynamic, realistic user feedback and simulate multi-turn interactions, providing a scalable alternative to live experiments for evaluating how well models

adapt to evolving user states [167; 166]. These diverse approaches collectively underscore a shift from static, ID-based modeling toward dynamic, semantically grounded, and multi-faceted representations of user intent.

## Integrating World Knowledge and Denoising Interactions

The integration of Large Language Models (LLMs) into user behavior modeling represents a paradigm shift from relying solely on sparse, ID-based collaborative signals to leveraging extensive external world knowledge. This transition addresses the fundamental limitation of traditional sequential recommendation systems, which often struggle with data sparsity and the inability to capture the semantic nuances of user intent. By incorporating world knowledge, frameworks can compensate for insufficient interaction data and provide a more holistic understanding of user dynamics. For instance, the GRASP framework exemplifies this approach by integrating generation-augmented retrieval with holistic attention mechanisms, effectively utilizing LLM-generated world knowledge to mitigate noise and capture dynamic user interests even in the presence of hallucinations [15]. Similarly, LWGR advances this integration by employing Lagrangian constraints to transfer personalized world knowledge from LLMs into generative recommendation, ensuring that the fusion of external knowledge does not conflict with behavioral signals while capturing multidimensional user heterogeneity [206].

A critical challenge in leveraging LLMs for lifelong behavior modeling is the "lost-in-the-middle" phenomenon and the difficulty LLMs face in comprehending long textual sequences of user interactions, even when these sequences fall within the model's context window limits. To address this, researchers have developed hierarchical and chunked processing strategies. The HiT-LBM framework proposes a hierarchical tree search-based approach that divides lifelong behaviors into manageable chunks, using a cascading extraction method combined with process rating models to ensure the quality of interest generation and mitigate error accumulation [218]. Complementing this, the LIBER framework utilizes a streaming partition strategy to condense lengthy sequences into shorter segments, allowing LLMs to infer insights in a cascading manner before fusing them into a comprehensive representation [200]. Furthermore, ReLLaX identifies this "lifelong sequential behavior incomprehension" as a distinct problem and offers a full-stack optimization solution involving semantic user behavior retrieval, soft prompt augmentation, and component fully-interactive LoRA to enhance the model's ability to process long sequences [24].

Beyond merely processing longer sequences, effective user modeling requires denoising interactions to distinguish genuine preferences from noisy or incidental clicks. LLMs offer unique reasoning capabilities to identify and filter such noise. The LLaRD framework leverages LLMs to generate denoising-related knowledge by enriching semantic insights and employing Chain-of-Thought techniques over user-item interaction graphs, subsequently using the Information Bottleneck principle to align this knowledge with recommendation targets [34]. In a different approach, ReaSeq utilizes explicit Chain-of-Thought reasoning via multi-agent collaboration to distill structured product knowledge, thereby combating the "knowledge poverty" of ID-based representations, while simultaneously employing generative behavior reasoning to infer plausible beyond-log behaviors, effectively expanding the signal space beyond observed interactions [109]. This generative capability is further explored in LLMDiRec, which integrates LLM-based semantics into an intent-aware diffusion model to preserve intent coherence and prevent the disruption of semantic consistency often caused by random data augmentation in contrastive learning [290].

The synthesis of world knowledge also extends to handling heterogeneous and cross-domain behaviors, where user interests may manifest differently across various contexts. The HUM method addresses the domain seesaw phenomenon and noise in heterogeneous behaviors by employing a compression enhancer and a robustness enhancer, allowing LLMs to effectively compress and understand diverse user interactions [204]. Similarly, TEMA-LLM utilizes LLMs to generate descriptive tags that enrich item representations, enabling a multi-attention mechanism to jointly model user preferences within and across domains [230]. For scenarios where real-time inference latency is a concern, frameworks like REKI and PersonaX decouple the knowledge infusion process from online inference. REKI extracts and condenses user and item knowledge offline into augmented vectors compatible with conventional models, while PersonaX segments historical behaviors into clustered groups to generate cached textual personas, significantly reducing online latency while preserving diverse user interests [40] [54].

Despite these advancements, contradictions and gaps remain regarding the optimal balance between collaborative signals and semantic world knowledge. While some studies suggest that direct alignment of LLM representations with collaborative models is suboptimal due to information theoretic constraints, advocating instead for disentangled alignment [129], others argue for tight integration through hybrid prompting or dual-view modeling to maximize the benefits of both modalities [146] [290]. Furthermore, the reliance on LLMs introduces concerns regarding hallucination

and the potential for generated knowledge to misalign with actual user behavior, necessitating robust denoising and verification mechanisms such as those found in GRASP and LWGR. The field is increasingly moving towards foundation models specifically tailored for behavior, such as BehaveGPT, which utilizes a DRO-based pretraining paradigm to equitably model both head and tail behaviors, addressing the imbalance that general LLMs often fail to capture [142]. This evolution suggests a future where user behavior modeling is not just an application of general LLMs but a specialized domain requiring architectures that inherently understand the temporal, contextual, and semantic complexities of human decision-making.

## Conversational Interaction, Preference Elicitation, and Explainability

The evolution of recommender systems from static, retrieval-based paradigms to dynamic, generative frameworks has fundamentally altered the nature of user-system interaction. Traditional systems rely on passive feedback mechanisms such as clicks and views, which often fail to capture the nuanced and evolving information needs of users [195]. In contrast, generative recommendation leverages Large Language Models (LLMs) to facilitate active collaboration through natural language dialogue, transforming the user experience into an interactive process where preferences are elicited, refined, and explained in real-time [69]. This shift enables systems to move beyond merely ranking existing items to engaging in multi-turn conversations that clarify user intent, generate personalized content, and provide transparent justifications for recommendations [92].

A central component of this interactive paradigm is effective preference elicitation. While traditional conversational recommender systems (CRS) often rely on explicit questions regarding item attributes, this approach assumes a level of domain knowledge that many users lack [Generating Usage-related Questions for Preference Elicitation in Conversational Recommender Systems]. To address this, recent research proposes generating implicit, usage-related questions that inquire about the context in which an item will be used rather than specific technical specifications [Generating Usage-related Questions for Preference Elicitation in Conversational Recommender Systems]. This method allows users to express needs based on their intended activities, which LLMs can then map to item attributes. Furthermore, the integration of LLMs enables the system to interpret complex, narrative-driven queries where users describe their preferences and constraints in verbose natural language, a capability that traditional models struggle to support without significant data augmentation [43]. By treating recommendation as an instruction-following task, systems can dynamically adjust to user feedback, steering recommendations based on natural language commands that specify what to include or exclude [18].

The fidelity of these interactions often depends on the system's ability to maintain context and understand sequential dependencies within a conversation. However, standard LLMs may fail to fully capture the sequential dynamics of user behavior, leading to recommendations that ignore the temporal evolution of preferences [145]. To mitigate this, frameworks like MemoCRS incorporate memory-enhanced mechanisms that store and retrieve relevant information from historical dialogue sessions, ensuring continuity and reducing noise in long-term interactions [131]. Additionally, automatic memory-retrieval frameworks have been developed to extract relevant long-term interests from extensive user histories, addressing the context window limitations of standard LLMs [13]. The challenge of evaluating these conversational agents has led to the development of sophisticated user simulators. Since collecting real-world conversational data is costly, generative reward models and LLM-based simulators are employed to create synthetic users capable of providing fine-grained feedback, such as attribute-based critiquing, thereby enabling robust multi-turn training and evaluation [203]. These simulators allow for the assessment of how well models emulate human behavior across diverse tasks, including expressing open-ended preferences and giving feedback [150].

Explainability serves as another critical pillar of generative conversational interaction, fostering trust and allowing users to scrutinize system logic. Unlike black-box collaborative filtering models, LLM-based systems can generate natural language explanations that articulate the reasoning behind a recommendation, linking item features directly to expressed user preferences [155]. Advanced frameworks integrate explanation generation directly into the recommendation pipeline, utilizing chain-of-thought reasoning to analyze user preferences and item perceptions before producing a final suggestion [80]. This capability extends to "scrutable" systems where users can interact with natural language profiles to understand and control how their data influences recommendations, thereby enhancing transparency and user agency [30]. Moreover, recent approaches combine sequence-aware modeling with utility-aware evaluation to ensure that generated explanations are not only linguistically coherent but also practically useful in influencing user decisions [189].

Despite these advancements, contradictions and challenges remain regarding the balance between conversational fluency and recommendation accuracy. Some studies indicate that while LLMs excel at generating explainable and diverse responses, they may lack the personalization depth of traditional models trained on extensive interaction logs, particularly in cold-start scenarios [296]. To bridge this gap, hybrid architectures are emerging that combine the semantic reasoning of LLMs with the collaborative signals of traditional models. For instance, retrieval-augmented generation (RAG) approaches allow LLMs to access external knowledge bases and interaction histories, grounding

their conversational responses in empirical data rather than relying solely on parametric knowledge [196]. Additionally, unified generative frameworks are being developed to handle both search and recommendation tasks within a single conversational flow, using dual-purpose identifiers to balance semantic relevance with collaborative filtering signals [265]. The ultimate goal of these integrations is to create systems that do not merely respond to queries but actively guide users through a discovery process, offering explanations, asking clarifying questions, and adapting to feedback in a manner that mimics human expert consultation [91].

## Multi-Turn Dialogue and Preference Steering

The integration of Large Language Models (LLMs) into Conversational Recommender Systems (CRS) has fundamentally shifted the paradigm from static, one-shot recommendations to dynamic, multi-turn interactions where users can actively steer the recommendation process through natural language. Unlike traditional systems that rely on implicit feedback such as clicks or purchases, LLM-enabled CRS facilitate complex dialogues that allow for nuanced preference elicitation, critiquing, and real-time adjustment of user intent [TITLE\_4]. This capability is particularly evident in e-commerce pre-sales scenarios, where the complementary strengths of LLMs' conversational fluency and traditional CRS's domain-specific accuracy are combined to better understand and elicit user needs [295]. The advent of datasets like REGEN, which includes natural language critiques and narratives, has provided a critical benchmark for training models to handle these steering queries, demonstrating that incorporating explicit user critiques significantly enhances recommendation quality by aligning language understanding with recommendation signals [14].

A central challenge in multi-turn dialogue is the effective tracking of user preferences across turns while managing the complexity of evolving intents. To address this, researchers have proposed frameworks like RA-Rec, which utilizes retrieval-augmented generation and prompt-based semi-structured state tracking to connect indirect user utterances, such as "I'm watching my weight," with complex item attributes found in reviews [196]. Furthermore, the continuity of preferences across historical and current sessions is vital for personalization. The MemoCRS framework addresses this by leveraging memory-enhanced LLMs that distinguish between user-specific memory for personalized interests and general memory for collaborative knowledge, thereby reducing noise in historical data and mitigating cold-start issues [131]. This approach contrasts with models that focus solely on the immediate dialogue session, highlighting the necessity of long-term context retention for effective preference steering.

The mechanism of preference steering often relies on the system's ability to generate appropriate responses or questions that guide the user toward clarifying their needs. While traditional preference elicitation asks direct questions about item attributes, which may require domain expertise from the user, novel approaches propose generating implicit, usage-related questions based on item reviews [173]. This method allows users to express preferences through intended usage scenarios rather than technical specifications, making the interaction more natural. Additionally, the use of LLM-based user simulators has emerged as a powerful tool for optimizing multi-turn interactions. Frameworks like SMTPO employ user simulator-guided multi-turn preference optimization, using reinforcement learning to align simulator-generated feedback with true user preferences, thus overcoming the scarcity of real-world conversational data and preventing error accumulation during multi-step reasoning [72]. Similarly, the GRSU framework utilizes a generative reward model to simulate user feedback, enabling automatic multi-turn interactions that refine the system's understanding of intricate preferences without requiring constant human intervention [203].

Despite these advancements, contradictions and limitations persist regarding the reliability of LLMs in maintaining accurate preference states over long dialogues. While LLMs excel at natural language understanding, they often struggle to leverage behavioral data effectively, a gap addressed by frameworks like CRAG that combine collaborative filtering with LLMs to improve item coverage and accuracy, particularly for newly released items [42]. Moreover, the fidelity of simulated users remains a concern; evaluations indicate that while LLMs can mimic human behavior, deviations in expressing open-ended preferences or providing consistent feedback can impact the generalization of the recommender [150]. To ensure realistic interactions, methods like Direct Preference Optimization (DPO) have been applied to refine text generation, ensuring that dialogue summaries and recommendation rationales remain rich in information and aligned with human conversational patterns [124].

The architecture of these systems is also evolving to support more granular control. The LUMEN framework, designed alongside the REGEN dataset, exemplifies a unified multi-task model where a single LLM backbone handles critiquing, retrieval, and narrative generation simultaneously [14]. This end-to-end approach contrasts with modular pipelines that separate dialogue management from recommendation engines, offering a more cohesive experience where each turn serves as an opportunity for both preference elicitation and explanation [20]. However, the reliance on LLMs introduces challenges in controlling the distribution of recommended items, leading to proposals like the Reindex-Then-Adapt (RTA) framework, which converts multi-token item titles into single tokens to better manage probability distributions during autoregressive generation [90]. Ultimately, the synergy between advanced dialogue management, memory

mechanisms, and retrieval-augmented strategies defines the current state of multi-turn preference steering, pushing CRS toward more transparent, controllable, and user-centric interactions.

## Conclusion

The integration of Large Language Models (LLMs) into recommender systems marks a definitive paradigm shift from traditional discriminative scoring to innovative generative frameworks. This transition fundamentally redefines the recommendation objective, moving away from estimating interaction probabilities within fixed candidate pools toward autoregressively generating items, identifiers, or narratives [4]. By leveraging the vast world knowledge, semantic reasoning, and natural language understanding inherent in LLMs, generative models address persistent limitations of historical approaches, including data sparsity, cold-start problems, and the lack of interpretability in ID-based embeddings [8]. The evolution from modular, multi-stage pipelines to unified generative architectures enables systems to operate effectively in zero-shot scenarios, bridge the gap between content semantics and collaborative signals, and facilitate interactive, conversational user experiences that were previously unattainable [51].

However, this paradigm shift introduces complex challenges that necessitate sophisticated architectural innovations and hybrid strategies. A central tension exists between the semantic fluency of LLMs and the statistical precision required for accurate collaborative filtering, often manifesting as hallucinations, distributional biases, and inefficiencies in autoregressive decoding [28]. To mitigate these issues, the field has converged on solutions that ground generative capabilities in factual data through retrieval-augmented generation, semantic identifier tokenization, and explicit alignment mechanisms that project item-level collaborative signals into the language model's token space [148]. Furthermore, theoretical analyses suggest that token-level generation cannot faithfully substitute for item-level generation without inducing distributional distortion, prompting frameworks like SimGR to model item-level preference distributions directly in a shared latent space [58]. Additionally, the computational demands of large-scale deployment have spurred the development of parameter-efficient fine-tuning, device-cloud collaboration, and speculative decoding techniques, ensuring that the rich reasoning capabilities of LLMs can be harnessed without compromising the low-latency requirements of industrial applications [Efficient Inference for Large Language Model-Based Generative Recommendation].

Looking forward, the trajectory of generative recommendation points toward the emergence of autonomous agents capable of dynamic, lifelong user modeling and proactive preference elicitation. Future systems will likely transcend passive retrieval to become active creators that can repurpose content, simulate user feedback for robust training, and provide transparent, narrative-driven explanations that foster trust [195]. Realizing this vision requires a holistic approach that balances the expressive power of world knowledge with the rigor of behavioral data, ensuring that next-generation recommenders are not only semantically rich and explainable but also behaviorally accurate, scalable, and ethically aligned with diverse user needs [51].